TATA	**FIR** **VEHICLE TESTING**	**28/07/2023**

FIR No :FIR/Testing-Outdoor/518000/2023/6354

Part No/ Group :51800151R

Subject :Performance evaluation of Signa 3121.T TATA 5L Cx BSVI-II vehicle in comparison with Ashok Leyland 3120 BSVI and Signa 3118.T Cummins 5.6L Ashwamedh BSVI-I vehicle.

Test Vehicle	Project Name :P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T Model Name : Signa 3121.T TATA 5L Cx BSVI-II	Enclosures : Annexure 1 - 5
	Chasis Number :PC13510122RJ00816 Engine Number :HXX104548	
FIR Status :	ESO FIR	

Role	Name	Signature
Prepared By	Palash Das	28-Jul-2023
Checked By	Swapnil Jadhav	28-Jul-2023
Approved By	Ajaykumar Jadhav	28-Jul-2023

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA 20 - F 222 / May19

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Documents : Please click here

FIR_Testing_Outdoor_51800..

Subject: -	Performance evaluation of Signa 3121.T TATA 5L Cx BSVI-II vehicle in comparison with Ashok Leyland 3120 BSVI and Signa 3118.T Cummins 5.6L Ashwamedh BSVI-I vehicle.		
Conclusion:			
➢ **Signa 3121.T 5L Cx BSVI-II in comparison with Ashok Leyland 3120 BSVI**			
• Overall performance of Signa 3121.T 5L Cx BSVI-II is better as compared to Ashok Leyland 3120 BSVI vehicle, PALS position "C" achieved against "C".		--	--
➢ **Signa 3121.T 5L Cx BSVI-II in comparison with Signa 3118.T Ash. BSVI-I**			
• Overall performance of Signa 3121.T 5L Cx BSVI-II is better as compared to Signa 3118.T Ash. BSVI-I base vehicle.		--	--
1.1 Acceleration:			
➢ **Signa 3121.T 5L Cx BSVI-II in comparison with Ashok Leyland 3120 BSVI**			
• Acceleration (0-60 kmph) of Signa 3121.T 5L Cx BSVI-II was found to be **51.03** Sec. Signa 3121.T 5L Cx BSVI-II is **8.60%** better as compared to Ashok Leyland 3120 BSVI vehicle and meets PALS target.		--	--
• Acceleration (0-1000m) of Signa 3121.T 5L Cx BSVI-II was found to be **76.71** Sec. Signa 3121.T 5L Cx BSVI-II is **1.78%** better as compared to Ashok Leyland 3120 BSVI vehicle and meets PALS target.		--	--
➢ **Signa 3121.T 5L Cx BSVI-II in comparison with Signa 3118.T Ash. BSVI-I**			
• Acceleration (0-60 kmph) of Signa 3121.T 5L Cx BSVI-II is **Superior by 6.66%** as compared to Signa 3118.T Ash. BSVI-I vehicle.		--	--
• Acceleration (0-1000m) of Signa 3121.T 5L Cx BSVI-II is **Superior by 7.48%** as compared to Signa 3118.T Ash. BSVI-I vehicle.		--	--
1.2 Wide open throttle drivability:			
➢ **Signa 3121.T 5L Cx BSVI-II in comparison with Ashok Leyland 3120 BSVI**			
• In gear drivability of Signa 3121.T 5L Cx BSVI-II vehicle was found to be **2.49** kmph/s, **2.85** kmph/s, **2.11** kmph/s, **1.4** kmph/s, **0.85** kmph/s, and **0.69** kmph/s in 1st , 2nd , 3rd , 4th , 5th and 6th gear respectively.		--	--

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• Rate of acceleration of Signa 3121.T 5L Cx BSVI-II vehicle is superior to Ashok Leyland 3120 BSVI in All gears.	--	--
1.3 <u>Bathtub drivability:</u>		
➢ **<u>Signa 3121.T 5L Cx BSVI-II in comparison with Ashok Leyland 3120 BSVI</u>**		
• Bathtub drivability of Signa 3121.T 5L Cx BSVI-II vehicle was found **better** as compared to Ashok Leyland 3120 BSVI vehicle in 4th, 5th & 6th gear and meets PALS target, except 3rd gear (25-30kmph) speed band.	VI/ENG	C
➢ **<u>Signa 3121.T 5L Cx BSVI-II in comparison with Signa 3118.T Ash. BSVI-I</u>**		
• Bathtub drivability of Signa 3121.T 5L Cx BSVI-II vehicle was found **superior** to Signa 3118.T Ash. BSVI-I vehicle in all gears.	--	--
1.4 <u>Sustain and Restart Gradability:</u>		
➢ **<u>Signa 3121.T 5L Cx BSVI-II in comparison with Ashok Leyland 3120 BSVI</u>**		
• Sustain gradeability of Signa 3121.T 5L Cx BSVI-II was found to be **27.28%.** Signa 3121.T 5L Cx BSVI-II is **17.08%** superior as compared to Ashok Leyland 3120 BSVI vehicle.	--	--
• Restart gradeability of Signa 3121.T 5L Cx BSVI-II was found to be **22.28%.** Signa 3121.T 5L Cx BSVI-II is **15.68%** superior as compared to Ashok Leyland 3120 BSVI vehicle.	--	--
➢ **<u>Signa 3121.T 5L Cx BSVI-II in comparison with Signa 3118.T Ash. BSVI-I</u>**		
• Sustain gradeability of Signa 3121.T 5L Cx BSVI-II is 5.74% superior as compared to Signa 3118.T Ash. BSVI-I base vehicle.	--	--
• Restart gradeability of Signa 3121.T 5L Cx BSVI-II is 4.50% superior as compared to Signa 3118.T Ash. BSVI-I base vehicle.	--	--
Note: Peformance Trial conducted on Ashok Leyland 2820 BSVI vehicle from which Ashok Leyland 3120 performance data is extraploted.		
2.0 <u>Result:</u> Wide observation.		

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FAILURES CLASS:

A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.

B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.

C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.

D: Customer perception issues and minor irritants.

3.0 Test Vehicle:

1. Signa 3121.T 5L Cx BSVI-II (ERC – 1269)
   Chassis No. – PC13510122RJ00816
   Engine No. – HXX104548

Input Doc Ref. : Testing-Outdoor/ 06442

DID No : NA

DDP No : NA

Enclosures:

Annexure 1: Panel chart

Annexure 2: PAT Sheet

Annexure 3: PAT Summary

Annexure 4: Test request

Annexure 5: Test plan

Work Order No. :- P.22.EA.501	**Make :-** Tata Motors Ltd	**T I No. – 42587**

Earlier Ref: -

- FIR_Testing Outdoor_508500_2021_3980_dated 17_08_2021_Performance evaluation on Ashok Leyland 2820 BS VI vehicle.
- FIR_Testing Outdoor_512600_2020_2859_dated 15_10_2020_Performance evaluation on Signa 3118.T Ash. BSVI-I Ashwamedh.

4.0 Project Status: DR3

5.0 Introduction:

- As a part of RDE program, Signa 3121.T 5L Cx BSVI-II vehicle was offered for performance evaluation, Ashok Leyland 3120 BSVI was taken as core benchmark and Signa 3118.T Ash. BSVI-I vehicle was taken as Base for performance comparison. This report will reveal detail of this trial with the status.

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6.0 Testing

➢ Following test was conducted during trials.

Sr. No.	Test Description	Standard Referred	Instrument used
1	Acceleration	TST_VT_WG_0002	VBOX SX with GPS, Weighbridge (CRS Gate)
2	WOT Drivability	TST_VT_WG_0002	
3	Bath -Tub Drivability	TST_VT_WG_0261	
4	Gradability	TST_VT_WG_11	
5	Restart Gradability	TST_VT_WG_127	
6	Coastdown	TST_VT_WG_36	

7.0 Vehicle details:

Sr. No.	Parameters				Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
					SIGNA 3118.T 5.6L Ash BSVI -I	Ashok Leyland 3120 BSVI	SIGNA 3121.T 5L CX BSVI-II
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)				ERC-1187	--	ERC-1269
2	Vehicle GVW/GCW			kg	31000	31000	31000
3	Engine Details (Make/cc/Norms/no. of Cyl.)				Ashwamedh BSVI 5.6L	H-Series CRS BSVI	TML 5LNGE
	Engine Max Power				180 HP @ 2300 rpm	200 hp @ 2400 rpm	204 HP @ 2200 rpm
	Engine Max Torque				850Nm @ 1000-1600 rpm	700 Nm @ 1200-1900 rpm	850Nm @ 1000-1600rpm
	Engine Calibration Maturity Level				As per production released	As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)				Time Based (100 Hrs)	--	Time Based (110 Hrs)
5	Gear Box (make/drive/FGR)				G950-FGR - 9.36,DD	ZF 6S-850 Gearbox	G950-FGR - 9.36,DD
	Gear Ratios				1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1-9.36 , 2-5.20 , 3-3.04 , 4-1.83 , 5-1.32 , 6-1.0
6	RAR				4.88	5.57	5.12
7	Tyre Details				All 295/90R 20 (Bridgestone Make)	F1 - Apollo 295/90R 20 R1, R2 - Apollo 295/90R 20	All 295/90R 20 (Bridgestone Make)
8	LBSC	High & Low Engine Speed Breakpoint		RPM	1900rpm, 1900rpm	--	--
		Vehicle Mass Threshold		kg	31000kg	--	--
	VAM	Variable Acceleration 1, 2		kmph/sec	2.5 kmph/Sec, 2.5 kmph/Sec	--	--
		Variable Acceleration Speed 1, 2		kmph	10 kmph, 40 kmph	--	--
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)		kmph	55 kmph, 55 kmph	--	--
	RSG	Maximum Accelerator Vehicle Speed		kmph	80	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor		Balance	100%	--	--
		Torque Limit Maximum Adjustment Factor		Economy	100%	--	--
	PTP	Powertrain Protection (Enable / Disable)			Disable	Disable	Disable
		Torque Limit		Nm	NA	NA	NA
	USFE / Torque Table Revision (1/2/3/n)				--	--	--
9	Fan Details (type/make/blades)				24'' E- Viscous fan	--	24'' E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)				As per production released	As per production released	95% Cal Maturity

Vehicle Photographs			

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8.0 Torque table:-

8.3.1 : Base vehicle (Signa 3118.T 5.6L BS-VI OBD-I (180 HP))

- Economy mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	753	400	400	400	400	400	400	80	0
1.33	500	500	850	800	753	543	400	400	400	400	80	0
1.84	500	500	850	800	753	543	400	400	400	400	80	0
3.05	500	500	850	800	800	753	543	400	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

- Balance mode

	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800
1	500	500	850	850	753	543	400	400	400	400	80	0
1.33	500	500	850	850	800	753	543	400	400	400	80	0
1.84	500	500	850	850	800	753	543	400	400	400	80	0
3.05	500	500	850	850	800	800	753	543	400	400	80	0
5.2	450	500	850	850	850	850	772	694	616	412	80	0
9.36	450	500	850	850	850	850	772	694	616	412	80	0

8.3.2: Test vehicle (Signa 3121.T 5L Cx BS-VI OBD-II (204 HP))

FE Switch Torque Table (Torque Limit)										
	Load Gea	1st Gear	2nd Gear	3rd Gear	4th Gear	5th Gear	6th Gear	7th Gear	8th Gear	Reverse
Economy Mode	NA	850	850	700	590	590	590	-	-	850
Balance Mode	NA	850	850	700	700	700	700	-	-	850
Power Mode	NA	850	850	850	850	850	850	-	-	850

9.0 Observation:-

Overall performance Signa 3121.T 5L Cx BSVI-II is better as compared to Ashok Leyland 3120 BSVI and PALS position achieved "C" against "C".

10. Future testing: - As per further development.

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Level 1 Attribute:

PAT- Performance and Drivability

PANEL chart

VE CVBU/ PD/001

Specification:
Max Power-204 hp @ 2200 rpm
Max Torque - 850 Nm 1000-1600 rpm
Driveline : G-950 DD | 5.12 RAR | 295/90R 20 Tyres | 3 Mode FE Switch | Torque Factor: NA
EOL Parameters : RSG: 80 kmph

Panel Chart No.: VE CVBU/ HCV/ Signa 3121.T 5L Cx BSVI-II / PD/01

Programme / MY : SIGNA 3121.T Cx 5L BSVI-II | **Gateway :** DR3 | **Owner: Ajaykumar Jadhav** | **Project Name: SIGNA 3121.T 5L BSVI-II (Cx Version)** | **Date:** 27/07/2023

Level 1 Summary:

		PALS	Status: Current/ Gateway	Status: Job#1 Predicted
PALS	**Level 1:**	**C**	**C**	

Statement Of Intent (SOI) or DNA Description:

Performance: Haulage application

Attribute Target Summary:

Level 2	Level 3	Level 4	Level 5	Level 6	PALS	Metric	Current Target: w.r.t Ashok Leyland 3120 BSVI	Status: Current/Gateway SIGNA 3121.T 5L Cx BSVI-II	Base Vehicle: Signa 3118.T Ash BSVI-I	Competitor Data: Core Benchmark Ashok Leyland 3120	Remarks
								Heavy mode	**Heavy mode**	--	
Specification								**LBSC - NA VAM- NA GDP- NA RSG - 80 kmph**	**LBSC - 1900,1900,31000 Kg VAM- 2.5 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 55 kmph, 55 kmph RSG - 80kmph**	--	--
Vehicle GVW (Payload)						kg		31000	31000	31000	--
Power (Max) @ rpm						HP		204 hp @ 2200 rpm	180 hp @ 2300 rpm	200 hp @ 2400 RPM	
Power to Weight ratio						HP / Ton		6.58	6.45	6.45	
Torque (Max) @ rpm						Nm		850 Nm @ 1000-1600 rpm	850 Nm @ 1000-1600 rpm	700 Nm at 1200-2000 rpm	
Torque to Weight ratio						Nm / Ton		27.42	27.42	22.58	
Tyres size						~		Bridgestone 295/90R20	Apollo 295/90 R20	Apollo JK Tyres 295/90 R20	
Gear Box Ratios						~		G950 DD, FGR -9.36	G950 DD, FGR -9.36	Eaton DD 6S Gearbox	
Final Drive Ratio						~		**5.12**	**4.88**	**5.57**	
Acceleration at rated load					**C**			**C**			
0-60 kmph						Sec	**≤ 55.83**	51.03	54.67	55.83	
0-1000 m						Sec	**≤ 78.10**	76.71	82.91	78.1	
Bathtub Drivability (Time) at Rated load					**C**			**C**			
3rd Gear											
15 - 20 Kmph						sec	**≤ 3.59**	2.45	3.38	3.59	
20 - 25 Kmph						sec	**≤ 2.75**	2.58	--	2.75	
25 - 30 Kmph						sec	**≤ 2.90**	4.37	--	2.90	
4th Gear											--
20 - 25 Kmph						sec	**≤ 5.04**	4.29	6.27	5.04	
25 - 30 Kmph						sec	**≤ 4.79**	3.91	4.40	4.79	
30 - 35 Kmph						sec	**≤ 4.31**	3.73	3.79	4.31	
5th Gear											
30-40 kmph						sec	**≤ 14.20**	11.03	12.77	14.20	
40-50 kmph						sec	**≤ 14.11**	10.70	11.81	14.11	
50-60 kmph						sec	**≤ 15.66**	13.50	--	15.66	
6th Gear											
30-40 kmph						sec	**≤ 19.31**	19.19	30.73	19.31	
40-50 kmph						sec	**≤ 19.57**	14.65	17.48	19.57	
50-60 kmph						sec	**≤ 19.67**	15.42	16.49	19.67	
60-70 kmph						sec	**≤ 22.28**	17.23	19.03	22.28	
Gradeability at rated load					**C**			**C**			
Sustain Gradeability						%	**≥ 23.30**	27.28	25.08	23.30	--
Restart Gradeability						%	**≥ 19.26**	22.28	20.81	19.26	

Executive Summary:

1. Ashok Leyland 3120.T BSVI have been taken for target setting.
2. Signa 3121.T 5L Cx BSVI-II was having 95% engine calibration maturation.
3. Ashok Leyland 3120 BSVI data is extraploted from Ashok Leyland 2820 BSVI data.
4.Target has been taken at-par as per Cx variant approach.

Tested by:

Palash Das / Swapnil Jadhav
PAT Engineer

Verified and Approved by:

Ajaykumar Jadhav / Sanjeev Kumar
PAT Lead / Project Lead

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PAT ATTRIBUTES SHEET
Performance and Drivability | Vehicle-level Engine Calibration
Features / End of Line | OBDII and Diagnostics

Document Version: VE CVBU/PD/001

Document Revision No.: VE CVBU/ HCV/ Signa 3121.T Cx BSVI-II/PD/ 001

Department Name: Vehicle Engineering

Project Name: Signa 3121.T 5L Cx BSVI-II
Base Vehicle- Signa 3118.T BSVI_G950 6S DD (FGR- 9.36)_RAR 4.88_295/90R20 tires_31Ton GVW_3 Mode FE Switches
Competitor Vehicle- Ashok Leyland 3120 BSVI_ZF 6S-850 DD (FGR- 8.51)_RAR 5.57_295/90R20 tires_31Ton GVW
Application- Haulage

Date: 27/07/2023

Sr.No.	Aggregate and test parameter: Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Unit	Test Type	Classification @DR3	Ref.Standard / Work Guideline	Simulation	Base Vehicle: Signa 3118.T BSVI-I G950 DD 6S RAR 4.88	Core Benchmark: Ashok Leyland 2820 Eaton 6 Speed DD RAR 5.57	Core Benchmark: Ashok Leyland 3120 Eaton 6 Speed DD RAR 5.57(Extraploted)	Target	Test Vehicle: Signa 3121.T 5L Cx BSVI-II G950 DD RAR 5.12	Status (RYG)	Remarks	CoC Remark
A	Performance Evaluation								Heading			LBSC - 1900,1900,31000 Kg VAM- 2.5 Kmph/s, 2.5 Kmph/s, 10 kmph, 40kmph GDP- 55 kmph, 55 kmph RSG - 80kmph	--	--	--	LBSC - NA VAM- NA GDP- NA RSG - 80 kmph	--	--	
1.1	Minimum speed								Heading			Heavy Mode				Heavy Mode	--		
1.1.1			1st gear				Kmph	Standard	Specification	TST/VT/WG/02		3.2	2.7	2.4	--	2.7	--	--	
1.1.2			2nd gear				Kmph		Specification			5.8	5.3	4.8		4.8	--	--	
1.1.3			3rd gear				Kmph		Specification			10.1	8.6	7.8		8.7	--	--	
1.1.4			4th gear				Kmph		Specification			16.4	12.8	11.6		14.9	--	--	
1.1.5			5th gear				Kmph		Specification			21.9	18.9	17.1		20.7	--	--	
1.1.6			6th gear				Kmph		Specification			29.4	23.8	21.5		27.3	--	--	
1.2	Maximum speed								Heading								--		
1.2.1			1st gear				Kmph	Standard	Specification	TST/VT/WG/02		7.3	12.4	11.2	--	10.5	--	--	
1.2.2			2nd gear				Kmph		Specification			13.2	21.1	19.1		18.2	--	--	
1.2.3			3rd gear				Kmph		Specification			24.0	34.7	31.3		30.5	--	--	
1.2.4			4th gear				Kmph		Specification			42.5	51.7	46.7		49.5	--	--	
1.2.5			5th gear				Kmph		Specification			54.4	71.4	64.5		67.2	--	--	
1.2.6			6th gear				Kmph		Specification			80.0	75.0	67.7		80.9	--	--	
1.2.9			Vehicle maximum speed				Kmph		Specification			80	80.0	80	80	80	G	--	
1.4	Acceleration								Heading								--		
1.4.2		AC Off							Heading								--		
1.4.2.1			0-60 kmph (0-37 mph)				Sec	Standard	Target	TST/VT/WG/02		54.67	50.43	55.83	≤ 55.83	51.03	G		
1.4.2.4			0-1000 m				Sec		Target			76.50	75.84	83.97	≤ 78.10	75.30	G		
1.5	Grad ability								Heading								--		
1.5.2		AC Off							Heading								--		
1.5.2.1			Maximum (sustained) Gradeability (forward)				%	Standard	Target	TST/VT/WG/011		25.08	25.80	23.30	≥ 23.30	27.28	G	--	
1.5.2.2			Max Restart Gradeability (forward)				%	Standard	Target	TST/VT/WG/127		20.81	21.32	19.26	≥ 19.26	22.28	G		
B	Drivability Evaluation								Heading								--		
1.3	WOT Drivability (AC Off)								Heading								--		
1.3.1			1st gear 4-6 Kmph				Sec	Standard	Data Generation	TST/VT/WG/02		1.10	0.84	0.93	--	0.80	--	--	
1.3.2			2nd gear 8-12 Kmph				Sec		Data Generation			2.15	2.32	2.57	--	1.41	--		
1.3.3			3rd gear 12- 20 Kmph				Sec		Data Generation			5.33	6.04	6.69	--	3.79	--		
1.3.4			4th gear 19- 31 Kmph				Sec		Data Generation			10.86	5.69	6.30	--	8.54	--		
1.3.5			5th gear 25 - 42 Kmph				Sec		Data Generation			23.52	11.00	12.18	--	19.89	--		
1.3.6			6th gear 36 - 50 kmph				Sec		Data Generation			27.02	23.40	25.91	--	20.16	--		
1.3.10		AC Off (Rate of acceleration)							Heading								--		
1.3.11			1st gear 4-6 Kmph				Kmph/sec		Data Generation			1.82	2.38	2.15	--	2.49	--		
1.3.12			2nd gear 8-12 Kmph				Kmph/sec		Data Generation			1.86	1.72	1.55	--	2.85	--		
1.3.13			3rd gear 12- 20 Kmph				Kmph/sec		Data Generation			1.50	1.32	1.19	--	2.11	--		
1.3.14			4th gear 19- 31 Kmph				Kmph/sec		Data Generation			1.10	1.23	1.11	--	1.40	--		
1.3.15			5th gear 25 - 42 Kmph				Kmph/sec		Data Generation			0.72	0.82	0.74	--	0.85	--		
1.3.16			6th gear 36 - 50 kmph				Kmph/sec		Data Generation			0.52	0.60	0.54	--	0.69	--		
1.4	Bathtub Curve - Part Throttle Drivability (AC Off)								Heading								--		
1.4.13			3rd gear					Standard	Heading	TST/VT/WG/261								1. Ashok Leyland 3120.T BSVI which has been extraploted from Ashok Leyland 2820.T BSVI, have been taken for target setting. 2. Signa 3121.T Cx BSVI RDE was having 95% engine calibration maturation. 3.Target has been taken at-par as per Cx variant approach	
1.4.14			15 - 20 Kmph				sec		Balanced Target			3.38	3.24	3.59	≤ 3.59	2.45	G		
1.4.15			20 - 25 Kmph				sec		Balanced Target			--	2.48	2.75	≤ 2.75	2.58	G		
1.4.16			25 - 30 Kmph				sec		Balanced Target			--	2.62	2.90	≤ 2.90	4.37	R		
1.4.1			4th gear						Heading										
1.4.4			20 - 25 Kmph				sec		Balanced Target			6.27	4.55	5.04	≤ 5.04	4.29	G		
1.4.5			25 - 30 Kmph				sec		Target			4.40	4.33	4.79	≤ 4.79	3.91	G		
1.4.6			30 - 35 Kmph				sec		Target			3.79	3.89	4.31	≤ 4.31	3.73	G		
1.4.8			5th gear						Heading										
1.4.9			30-40 kmph				sec		Target			12.77	12.83	14.20	≤ 14.20	11.03	G		
1.4.10			40-50 kmph				sec		Target			11.81	12.74	14.11	≤ 14.11	10.70	G		
1.4.11			50-60 kmph				sec		Target			--	14.14	15.66	≤ 15.66	13.50	G		
1.4.13			6th gear						Heading										
1.4.14			30-40 kmph				sec		Balanced Target			30.73	17.44	19.31	≤ 19.31	19.19	G		
1.4.9			40-50 kmph				sec		Target			17.48	17.68	19.57	≤ 19.57	14.65	G		
1.4.15			50-60 kmph				sec		Target			16.49	17.77	19.67	≤ 19.67	15.42	G		
1.4.16			60-70 kmph				sec		Target			19.03	20.12	22.28	≤ 22.28	17.23	G		
C	Engine Calibration Activities								Heading								--		
1.0	System Verification Trials								Heading								--		
1.1		Powertrain Characteristics (AC ON)							Target				--	--	--	--	NA	--	
1.2.1		Powertrain Characteristics (AC OFF)							Target				--	--	Pass	Pass	G	--	
2.0	Vehicle Perfomance Trials								Heading								--		
2.1		Vehicle Speed detection							Target				--	--	Pass	Pass	G	--	
2.2		Engine Speed detection							Target				--	--	Pass	Pass	G	--	
2.3		Gear Detection							Target				--	--	Pass	Pass	G	--	
2.4		Driven Idle							Target				--	--	Pass	Pass	G	--	
3.0	Engine derate verification								Heading								--		
3.2		Limp home validation							Heading				--	--	Pass	Pass	G	--	
3.3		Reduced torque fault validation							Heading				--	--	Pass	Pass	G	--	
4.0	Evaluation of Take-off ability, Acceleration and Deceleration								Target						Pass		G		
D	Vehicle feature validation and Fuel Economy features / End of Line (EOL) parameters segreggation and verification								Heading								--		
1.0	Vehicle Mass Estimation								Target				--	--	Pass	Pass	NA	--	
2.0	Road Slope Estimation								Target				--	--	Pass	Pass	NA	--	
3.0	Vehicle acceleration management (VAM tuning)								Target				--	--	Pass	Pass	G	--	
4.0	Electronic vehicle speed Limiter (EVSL) / Load based speed control (LBSC)								Target				--	--	Pass	Pass	NA	--	
5.0	Vehicle speed limiter (Top and one-gear down)								Target				--	--	Pass	Pass	NA	--	
6.0	EVSL+ VAM / LBSC+VAM								Target				--	--	Pass	Pass	NA	--	
7.0	Gear Down Protection (GDP)								Target				--	--	Pass	Pass	NA	--	
8.0	Idle shutdown								Target				--	--	Pass	Pass	G	--	
9.0	Average fuel economy (AFE)								Target				--	--	Pass	Pass	G	--	
10.0	IFE								Target				--	--	Pass	Pass	NA	--	
11.0	Engine Brake control								Target				--	--	Pass	Pass	G	--	
12.0	Cruise control (CC)								Target				--	--	Pass	Pass	G	--	
13.0	Power Take off (PTO)								Target				--	--	Pass	Pass	NA	--	
14.0	Remote PTO								Target				--	--	Pass	Pass	NA	--	
15.0	Road Speed governor								Target				--	--	Pass	Pass	G	--	
16.0	Engine Protection shutdown								Target				--	--	Pass	Pass	G	--	
17.0	Powertrain protection								Target				--	--	Pass	Pass	NA	--	
18.0	Trip-Info								Target				--	--	Pass	Pass	G	--	
E	Environmental Trials Verification								Heading								--		
1.0	Summer trials								Heading								--		
1.4		Hot start							Target				--	--	Pass	Pass	G	Carry over from OBD I validation	
1.5		Low idle governor calibration							Target				--	--	Pass	Pass	G		
1.6		Driven Idle							Target				--	--	Pass	Pass	G		
1.7		Subjective Driveability Assessment							Target				--	--	Pass	Pass	G		
2.0	Winter trials								Heading								--		
2.2		Cold Start (at Ambient as per Application requirement)							Target				--	--	Pass	Pass	G	Carry over from OBD I validation	
2.3		Smoke level (visually) during free acceleration after start							Target				--	--	Pass	Pass	G		
2.4		Subjective Driveability Assessment							Target				--	--	Pass	Pass	G		
6.0	High altitude trials/ calibration								Heading								--		
6.2		Low idle governor calibration					rpm	New Devlopment	Target				--	--	Pass	Pass	G	Carry over from OBD I validation	
6.9.1		% Deterioration in Drivability with base calibration					%		Target				--	--	Pass	Pass	G		
6.10.1		Turbocharger duty cycle mapping							Target				--	--	Pass	Pass	G		
6.10.1.1		Smoke during driving present or not						New Devlopment	Target				--	--	Pass	Pass	G		
6.10.1.2		Assessment of turbocharger Whistling noise							Target				--	--	Pass	Pass	G		
6.11.1		Power calculation					%		Target				--	--	Pass	Pass	G		
F	Onboard Diagnostics								Heading								--		
5.0	OBD Diagnostic - TML tool verification								Heading				--	--	Pass	Pass	G	Carry over from OBD I validation	
7.0	Derate Strategies for Critical Faults								Heading				--	--	Pass	Pass	G		
8.0	Verification of OBD interface for Comprehensive Component Monitors								Heading				--	--	Pass	Pass	G		

Sign Off

Responsibility	Name	Signature
Prepared by: PAT Engineer	Palash Das/Swapnil	
Checked & Approved by: PAT Lead / Project Lead	Ajaykumar Jadhav / Sanjeev kumar	

Index for Status

- Off target, no resolution
- Marginally off target, resolution planned/low risk
- Met target, no concern

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PED evaluation of Signa 3121.T TATA 5L Cx BSVI-II in comparison with Ashok Leyland 3120 BSVI and Signa 3118.T Cummins 5.6L Ashwamedh BSVI-I Vehicle.

PED| 25.07.23

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

Content

TATA MOTORS
Connecting Aspirations

- ✓ Vehicle specification comparison
- ✓ Performance Data Summary
- ✓ Minimum and Maximum Gear Speed
- ✓ Wide open throttle performance
- ✓ Tractive effort comparison
- ✓ Acceleration
- ✓ WOT drivability
- ✓ Bathtub drivability
- ✓ Coast Down comparison
- ✓ Sustain and restart Gradeability

Vehicle Specification Comparison

TATA MOTORS
Connecting Aspirations

Sr. No.	Parameters			Base Vehicle : OBD-1	Core Benchmark	Test Vehicle : OBD-2
				SIGNA 3118.T 5.6L Ash BSVI -I	Ashok Leyland 3120 BSVI	SIGNA 3121.T 5L CX BSVI-II
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1187	--	ERC-1248
2	Vehicle GVW/GCW		kg	31000	31000	31000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Ashwamedh BSVI 5.6L	H-Series CRS BSVI	TML 5LNGE
	Engine Max Power			180 HP @ 2300 rpm	200 hp @ 2400 rpm	204 HP @ 2200 rpm
	Engine Max Torque			850Nm @ 1000-1600 rpm	700 Nm @ 1200-1900 rpm	850Nm @ 1000-1600rpm
	Engine Calibration Maturity Level			As per production released	As per production released	95% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)	--	Time Based (110 Hrs)
5	Gear Box (make/drive/FGR)			G950-FGR - 9.36,DD	ZF 6S-850 Gearbox	G950-FGR - 9.36,DD
	Gear Ratios			1-9.36, 2- 5.20, 3- 3.05, 4- 1.84, 5- 1.33, 6- 1.00	1st: 8.51, 2nd: 4.66, 3rd: 2.73, 4th: 1.78, 5th: 1.27, 6th: 1, Rev:7.64	1-9.36 , 2-5.20 , 3-3.04 , 4-1.83 , 5-1.32 , 6-1.0
6	RAR			4.88	5.57	5.12
7	Tyre Details			All 295/90R 20 (Bridgestone Make)	F1 - Apollo 295/90R 20 - Apollo 295/90R 20	All 295/90R 20 (Bridgestone Make)
8	LBSC	High & Low Engine Speed Breakpoint	RPM	1900rpm, 1900rpm	--	--
		Vehicle Mass Threshold	kg	31000kg	--	--
	VAM	Variable Acceleration 1, 2	kmph/sec	2.5 kmph/Sec, 2.5 kmph/Sec	--	--
		Variable Acceleration Speed 1, 2	kmph	10 kmph, 40 kmph	--	--
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	55 kmph, 55 kmph	--	--
	RSG	Maximum Accelerator Vehicle Speed	kmph	80	80	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	100%	--	--
		Torque Limit Maximum Adjustment Factor	Economy	100%	--	--
	PTP	Powertrain Protection (Enable / Disable)		Disable	Disable	Disable
		Torque Limit	Nm	NA	NA	NA
	USFE / Torque Table Revision (1/2/3/n)			--	--	--
9	Fan Details (type/make/blades)			24'' E- Viscous fan	--	24'' E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)			As per production released	As per production released	95% Cal Maturity

Performance Data Summary

TATA MOTORS
Connecting Aspirations

Parameter	Unit	Target	Intended PALS	Benchmark	Base Vehicle	Test Vehicle			RDE Status (%) w.r.t Target	RDE Status (%) w.r.t Base	Remarks
				Ashok Leyland 3120 BSVI 6S DD RAR 5.57	Signa 3118.T BSVI_HM RAR 4.88	Signa 3121.T 5L BSVI RDE LM RAR 5.12	Signa 3121.T 5L BSVI RDE BM RAR 5.12	Signa 3121.T 5L BSVI RDE HM RAR 5.12			
Accleration											
0 - 60 kmph	sec	**≤ 55.83**	C	55.83	54.67	62.74	59.2	**51.03**	8.60%	6.66%	
0 - 1000 m	sec	**≤ 78.10**		83.97	82.91	81.78	79.76	**76.71**	1.78%	7.48%	
Driveability											
1st Gear(4-6)	sec	Not Applicable		0.93	1.10	0.86	0.85	**0.80**	Target Not available	Target Not available	
2nd Gear(8-12)	sec			2.57	2.15	1.47	1.40	**1.41**			
3rd Gear(12-20)	sec			6.69	5.33	4.43	4.39	**3.79**			
4th Gear(19-31)	sec			6.30	10.86	12.93	10.55	**8.54**			
5th Gear(25-42)	sec			12.18	23.52	27.06	22.06	**19.89**			
6th Gear(36-50)	sec			25.91	27.02	34.84	26.64	**20.16**			
Rate of acceleration											
1st Gear(4-6)	kmph/sec	Not Applicable		2.15	1.82	2.31	2.35	**2.49**	Target Not available	Target Not available	
2nd Gear(8-12)	kmph/sec			1.55	1.86	2.73	2.85	**2.86**			
3rd Gear(12-20)	kmph/sec			1.19	1.50	1.80	1.82	**2.11**			
4th Gear(19-31)	kmph/sec			1.11	1.10	0.93	1.14	**1.40**			
5th Gear(25-42)	kmph/sec			0.74	0.72	0.63	0.77	**0.85**			
6th Gear(36-50)	kmph/sec			0.54	0.52	0.40	0.53	**0.69**			
Bath-Tub Driveability											
3rd Gear											
15-20	Sec	**≤ 3.59**	C	3.59	3.38	2.91	2.76	**2.45**	31.85%	27.61%	
20-25	Sec	**≤ 2.75**	C	2.75	--	3.15	2.80	**2.58**	6.12%	--	
25-30	Sec	**≤ 2.90**	C	2.90	--	4.93	4.77	**4.37**	-50.63%	--	
4th Gear											
20-25	Sec	**≤ 5.04**	C	5.04	6.27	5.37	4.74	**4.29**	14.95%	31.63%	
25-30	Sec	**≤ 4.79**	C	4.79	4.40	5.44	4.44	**3.91**	18.30%	11.06%	
30-35	Sec	**≤ 4.31**	C	4.31	3.79	5.71	4.47	**3.73**	13.38%	1.50%	
5th Gear											
30-40	Sec	**≤ 14.20**	C	14.20	12.77	15.41	12.99	**11.03**	22.35%	13.65%	
40-50	Sec	**≤ 14.11**	C	14.11	11.81	16.83	13.45	**10.70**	24.16%	9.38%	
50-60	Sec	**≤ 15.66**	C	15.66	--	17.41	14.72	**13.50**	13.77%	--	
6th Gear											
30-40	Sec	**≤ 19.31**	C	19.31	30.73	28.48	21.31	**19.19**	0.63%	37.56%	
40-50	Sec	**≤ 19.57**	C	19.57	17.48	25.32	19.21	**14.65**	25.14%	16.19%	
50-60	Sec	**≤ 19.67**	C	19.67	16.49	26.28	20.44	**15.42**	21.63%	6.52%	
60-70	Sec	**≤ 22.28**	C	22.28	19.03	30.41	30.50	**17.23**	22.67%	9.47%	
Gradability and Restartability											
Sustain Gradability	%	**≥ 23.30**	C	23.30	25.08	--	--	**27.28**	17.08%	17.07%	
Restart Gradability	%	**≥ 19.26**	C	19.26	20.81	--	--	**22.28**	15.68%	15.70%	

Signa 3121.T 5L Cx BSVI-II Performance PALS position achieved "C" against "C"

Acceleration

- Acceleration of Signa 3121.T 5L Cx BSVI-II was found to be **better as compared to target** in 0-60 kmph speed and 0-1000M band

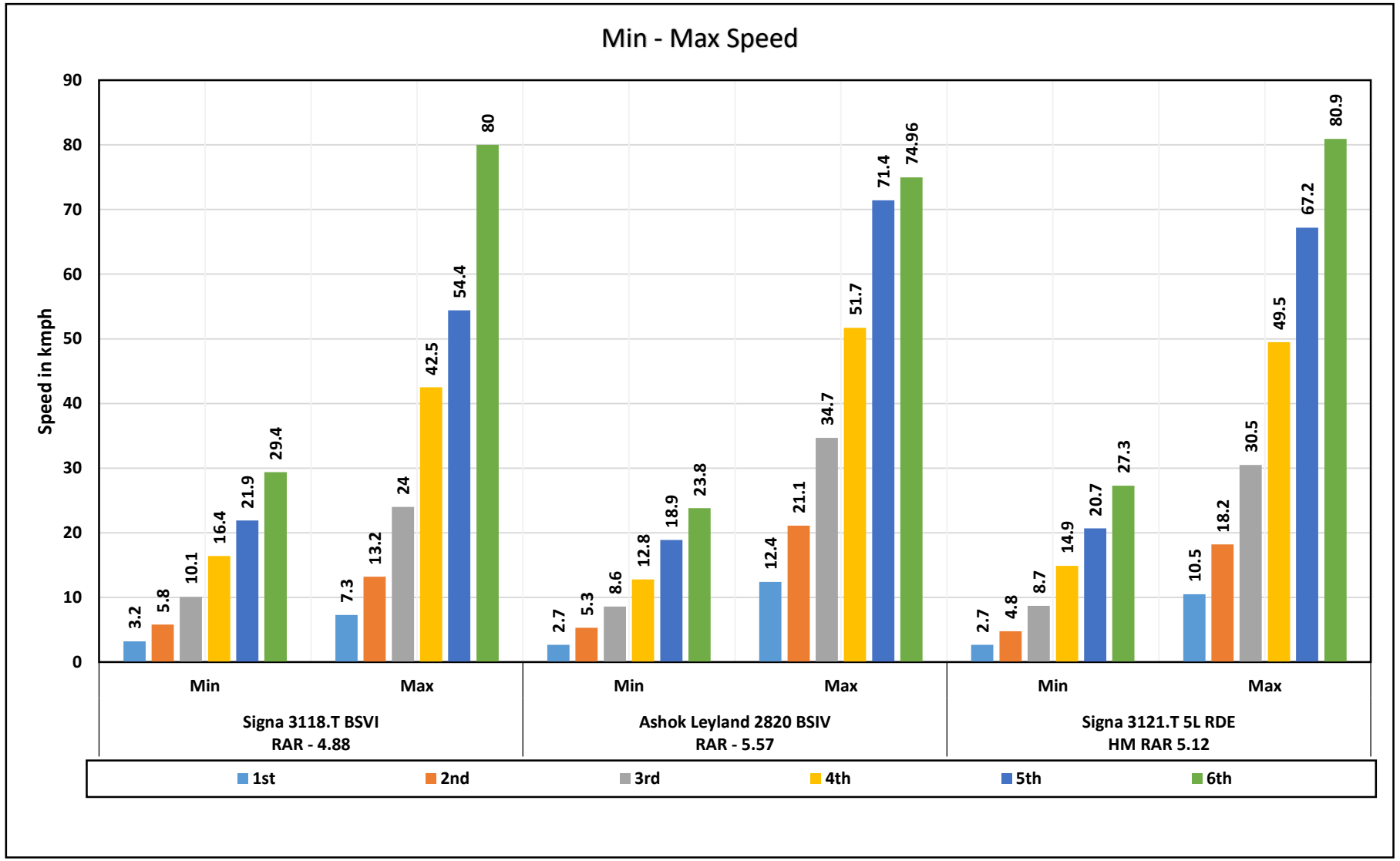
Bath tub Drivability

- Drivability of Signa 3121.T 5L Cx BSVI-II was found to be **better as compared** to target in all the speed bands except 3rd gear 25-30 kmph speed band.

Sustain & Restart Gradeability

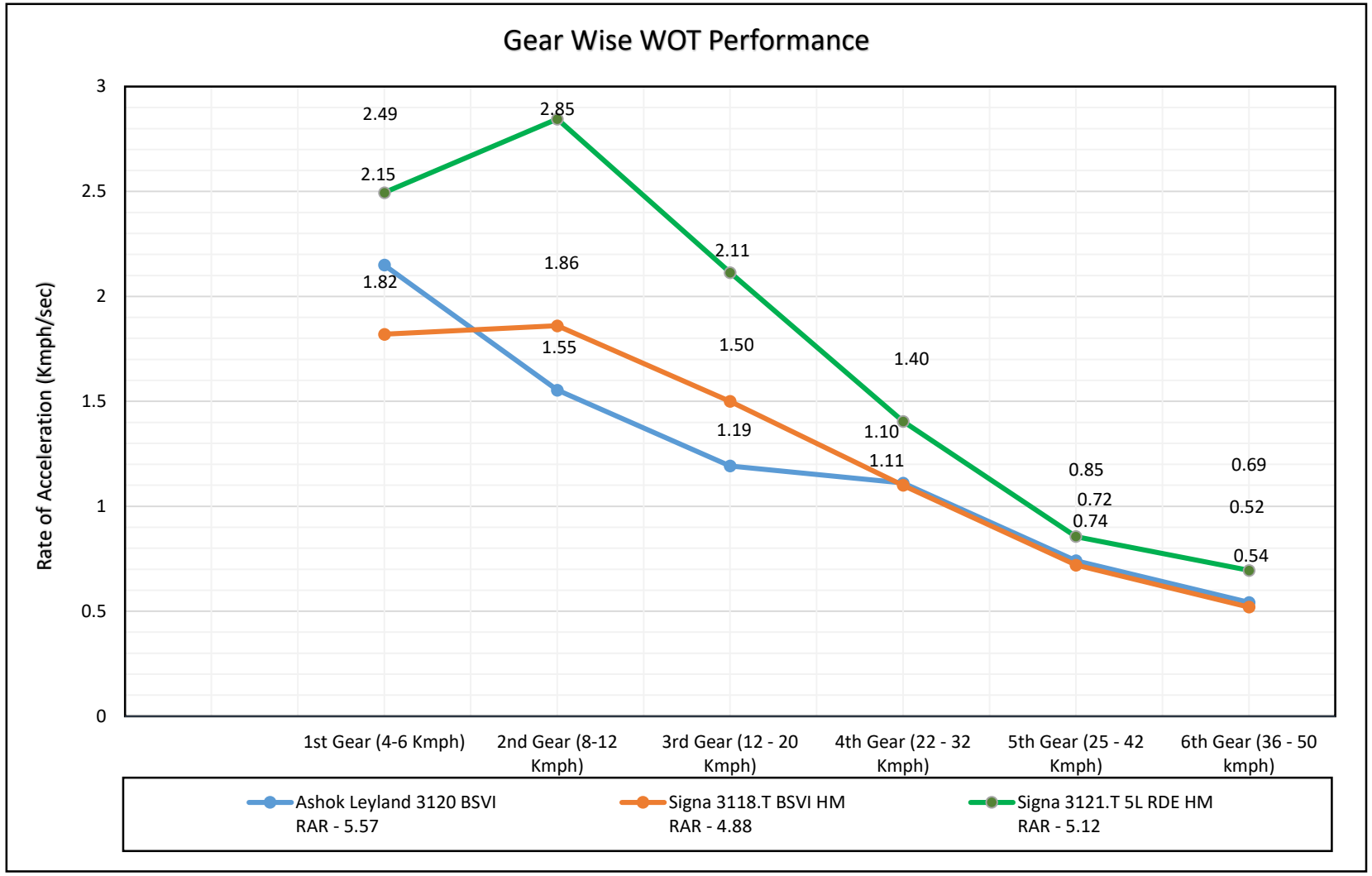
- Restart Gradability of Signa 3121.T 5L Cx BSVI-II was **found better** than target.
- Sustain Gradability of Signa 3121.T 5L Cx BSVI-II was **found better** that target

Minimum & Maximum Gear Wise speed Comparison



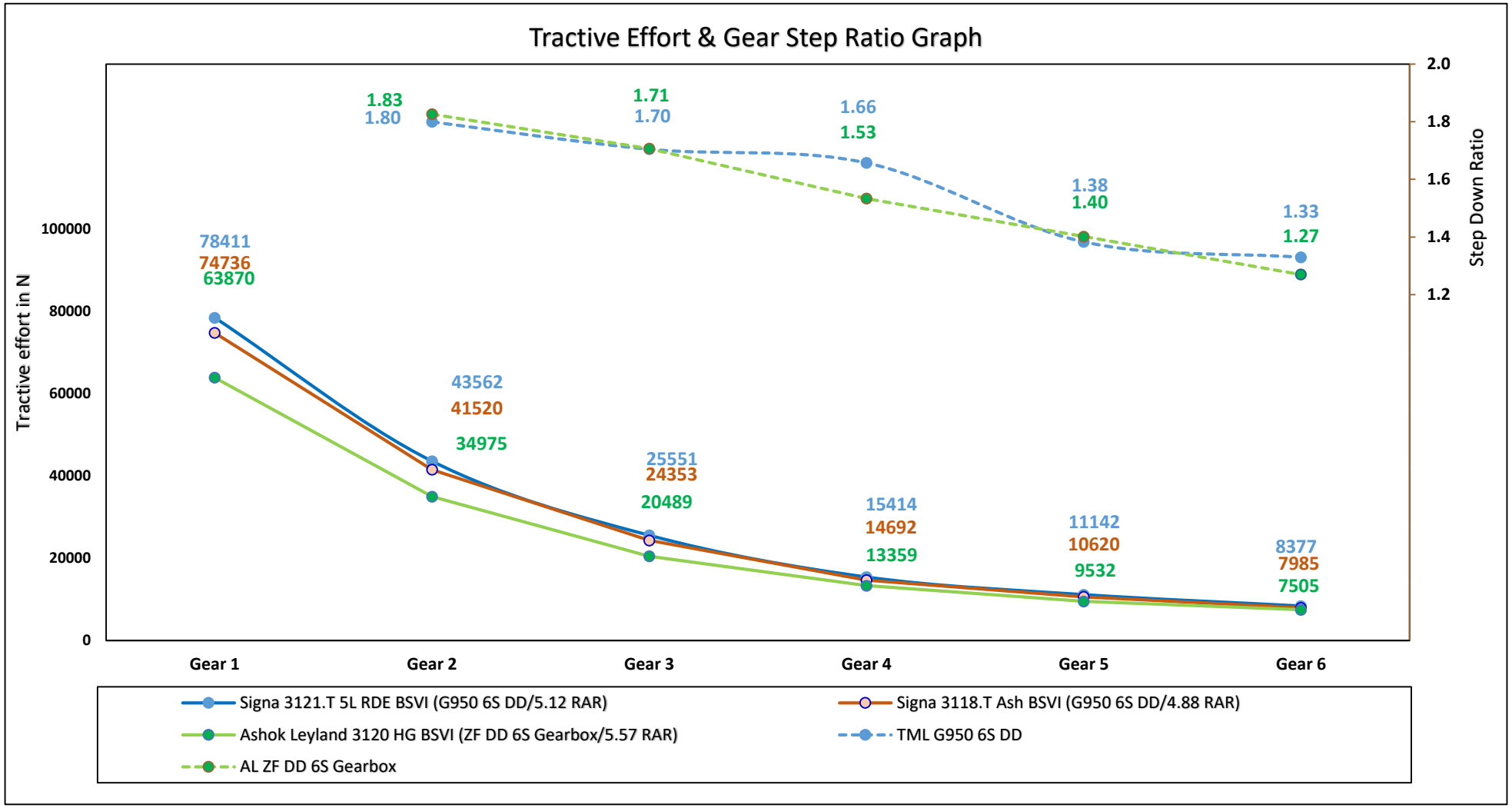
Gear	Signa 3118.T BSVI RAR - 4.88		Ashok Leyland 3120 BSIV RAR - 5.57		Signa 3121.T 5L RDE HM RAR 5.12	
	Min	Max	Min	Max	Min	Max
1st	3.2	7.3	2.7	12.4	2.7	10.5
2nd	5.8	13.2	5.3	21.1	4.8	18.2
3rd	10.1	24	8.6	34.7	8.7	30.5
4th	16.4	42.5	12.8	51.7	14.9	49.5
5th	21.9	54.4	18.9	71.4	20.7	67.2
6th	29.4	80	23.8	74.96	27.3	80.9

Wide open throttle Performance



Gear	Ashok Leyland 3120 BSVI RAR - 5.57	Signa 3118.T BSVI HM RAR - 4.88	Signa 3121.T 5L RDE HM RAR - 5.12
1st Gear (4-6 Kmph)	2.15	1.82	2.49
2nd Gear (8-12 Kmph)	1.55	1.86	2.85
3rd Gear (12 - 20 Kmph)	1.19	1.50	2.11
4th Gear (22 - 32 Kmph)	1.11	1.10	1.40
5th Gear (25 - 42 Kmph)	0.74	0.72	0.85
6th Gear (36 - 50 kmph)	0.54	0.52	0.69

Tractive effort



➢ Tractive effort of Signa 3121.T 5L Cx BSVI-II vehicle is better as compared with AL 3120 BSVI & Signa 3118.T Ash. BSVI-I in all gear.

Acceleration Performance – Through Gear

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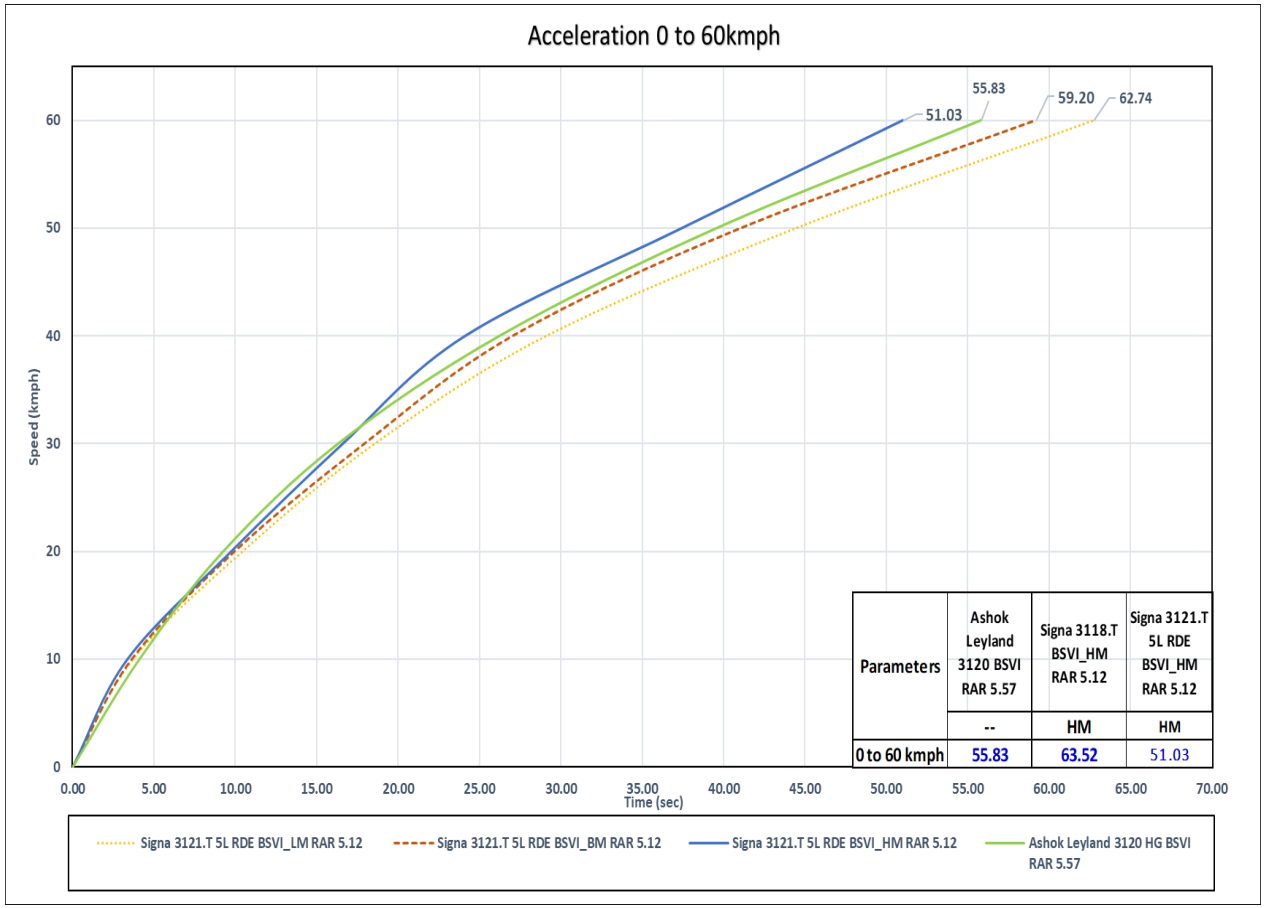
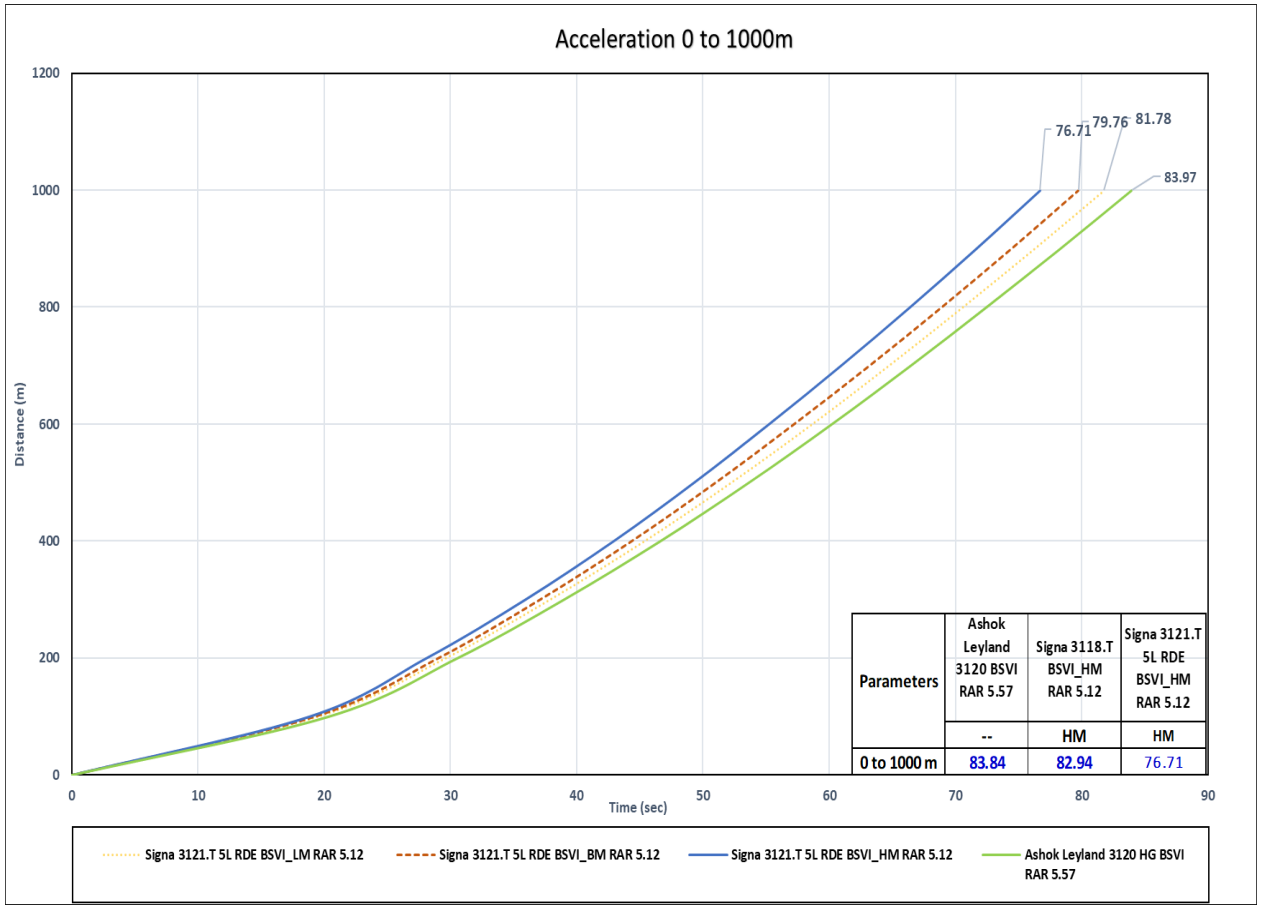


Parameters	Ashok Leyland 3120 BSVI RAR 5.57	Signa 3118.T BSVI_HM RAR 5.12	Signa 3121.T 5L RDE BSVI_HM RAR 5.12
	--	HM	HM
0 to 60 kmph	55.83	63.52	51.03



Parameters	Ashok Leyland 3120 BSVI RAR 5.57	Signa 3118.T BSVI_HM RAR 5.12	Signa 3121.T 5L RDE BSVI_HM RAR 5.12
	--	HM	HM
0 to 1000 m	83.84	82.94	76.71

➢ Acceleration (0-60kmph) & (0-1000M) , Signa 3121.T 5L Cx BSVI-II vehicle is better as compared with AL 3120 BSVI & Signa 3118.T Ash. BSVI-I .

WOT Drivability @ 1st & 2nd Gear

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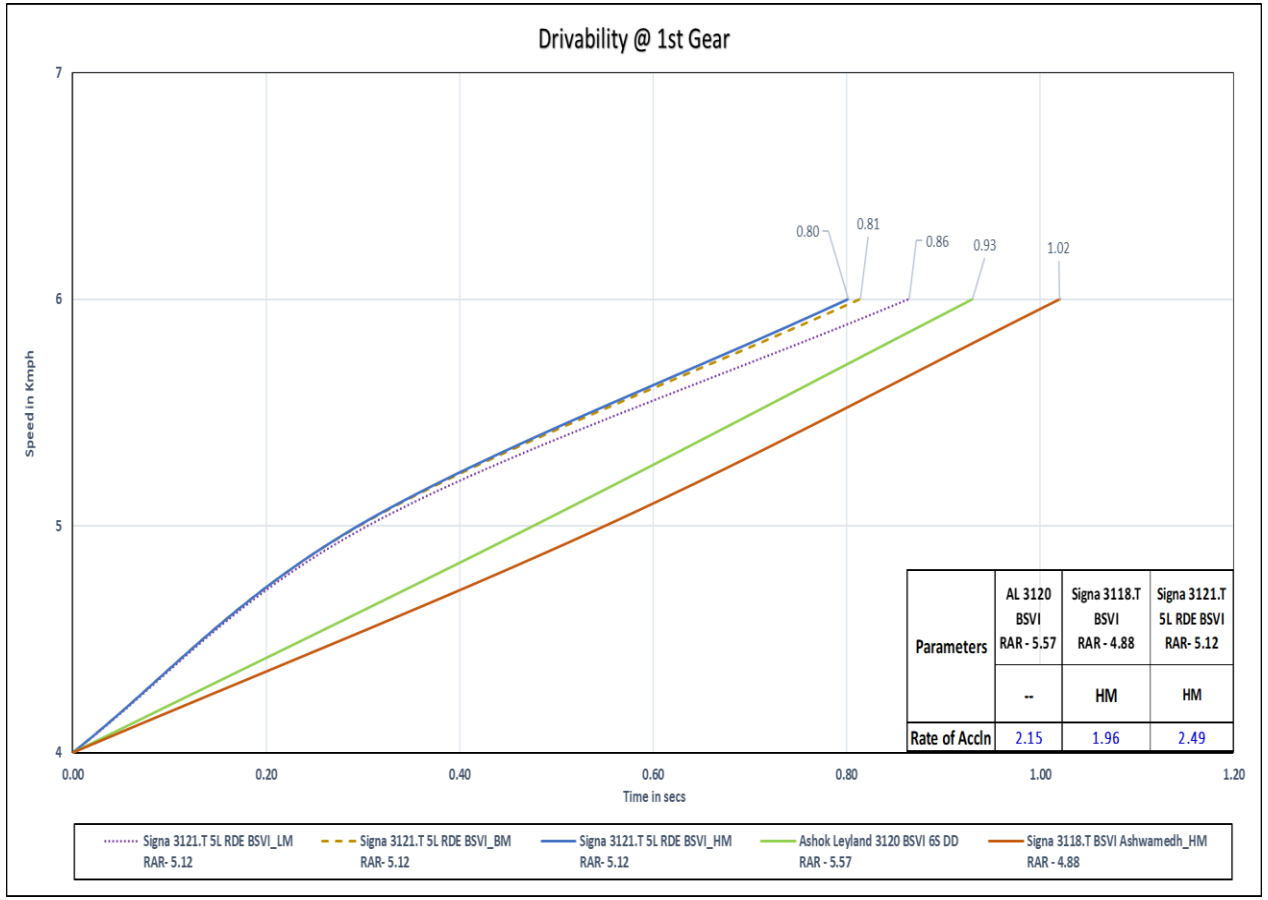
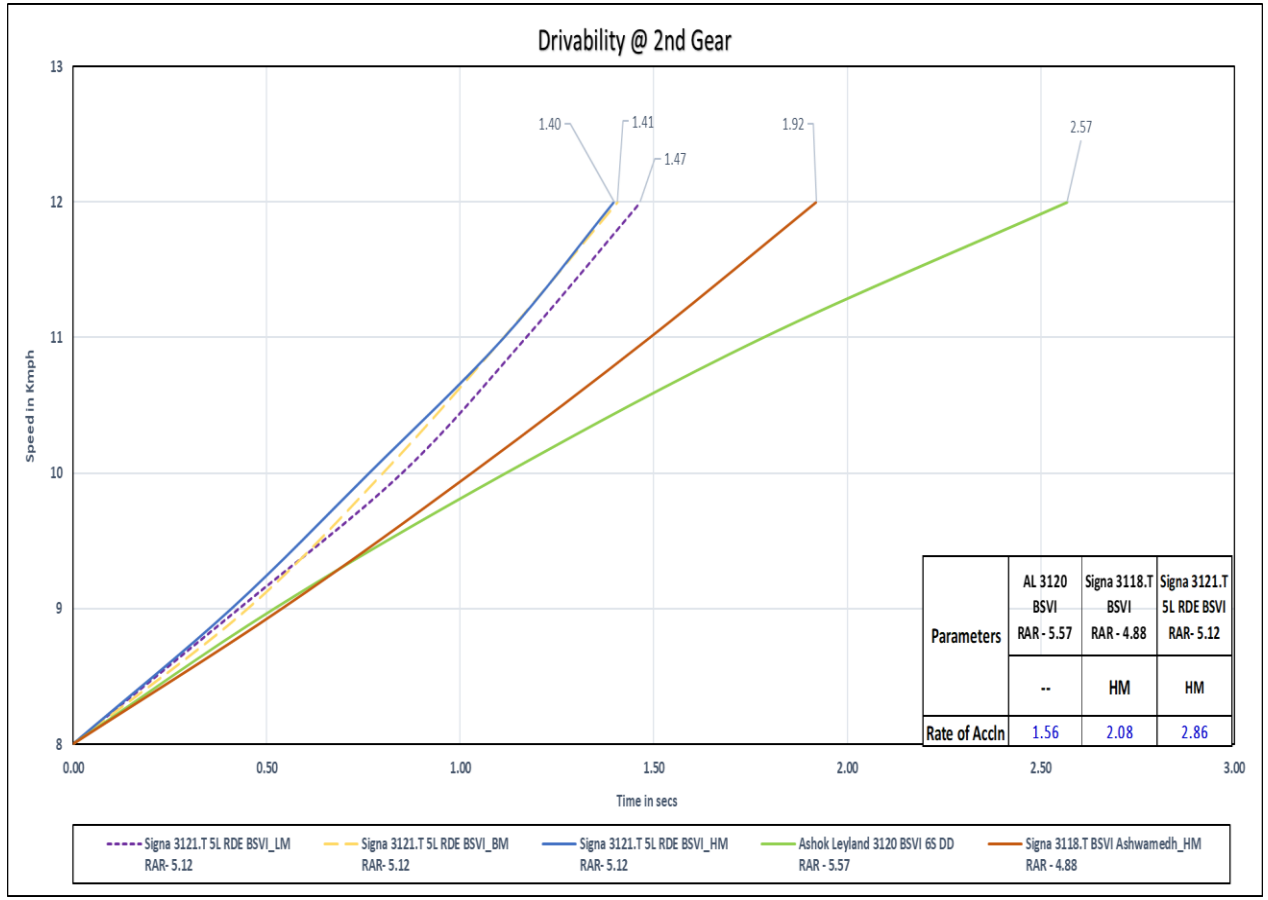
Connecting Aspirations



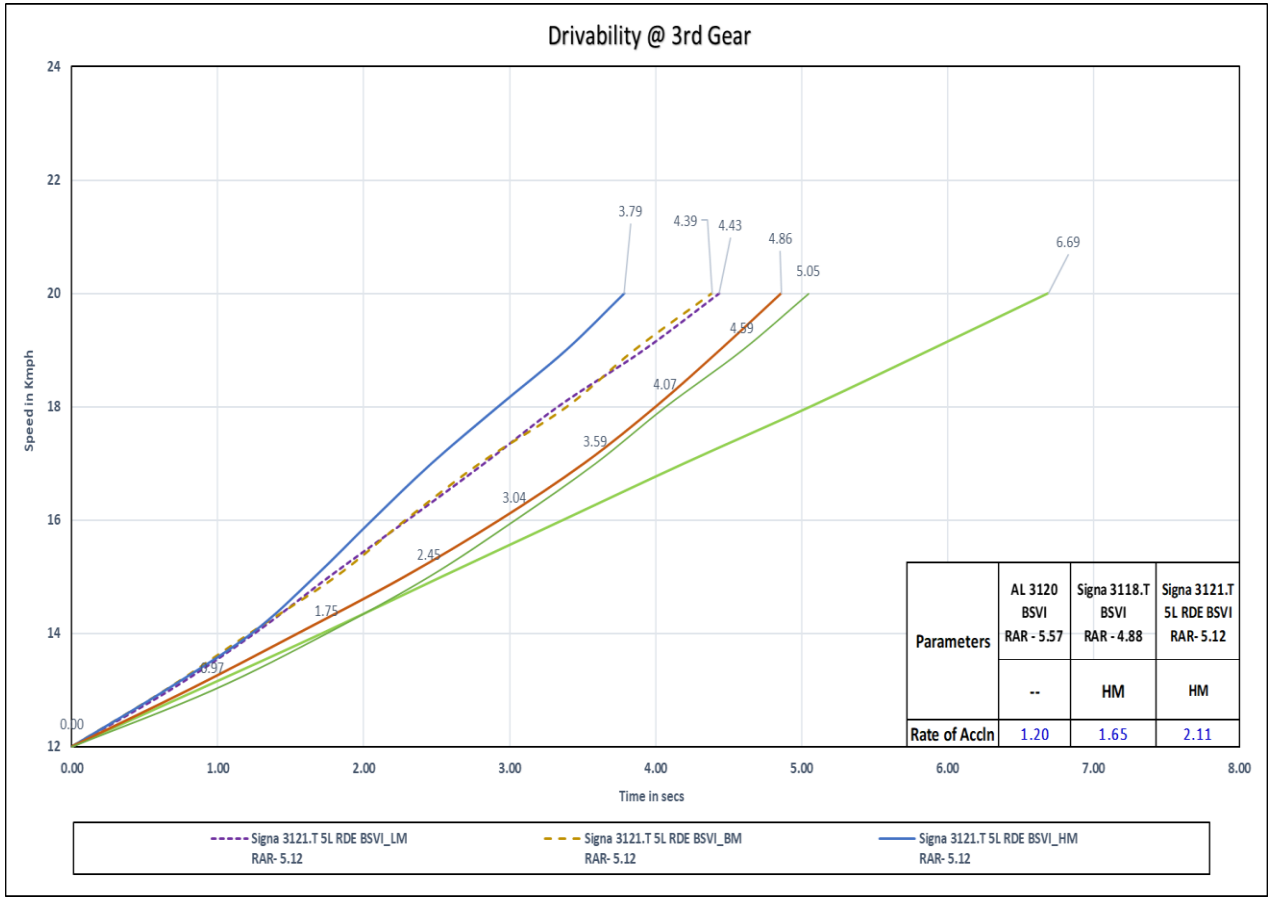
Parameters	AL 3120 BSVI RAR - 5.57	Signa 3118.T BSVI RAR - 4.88	Signa 3121.T 5L RDE BSVI RAR- 5.12
	--	HM	HM
Rate of Accln	2.15	1.96	2.49

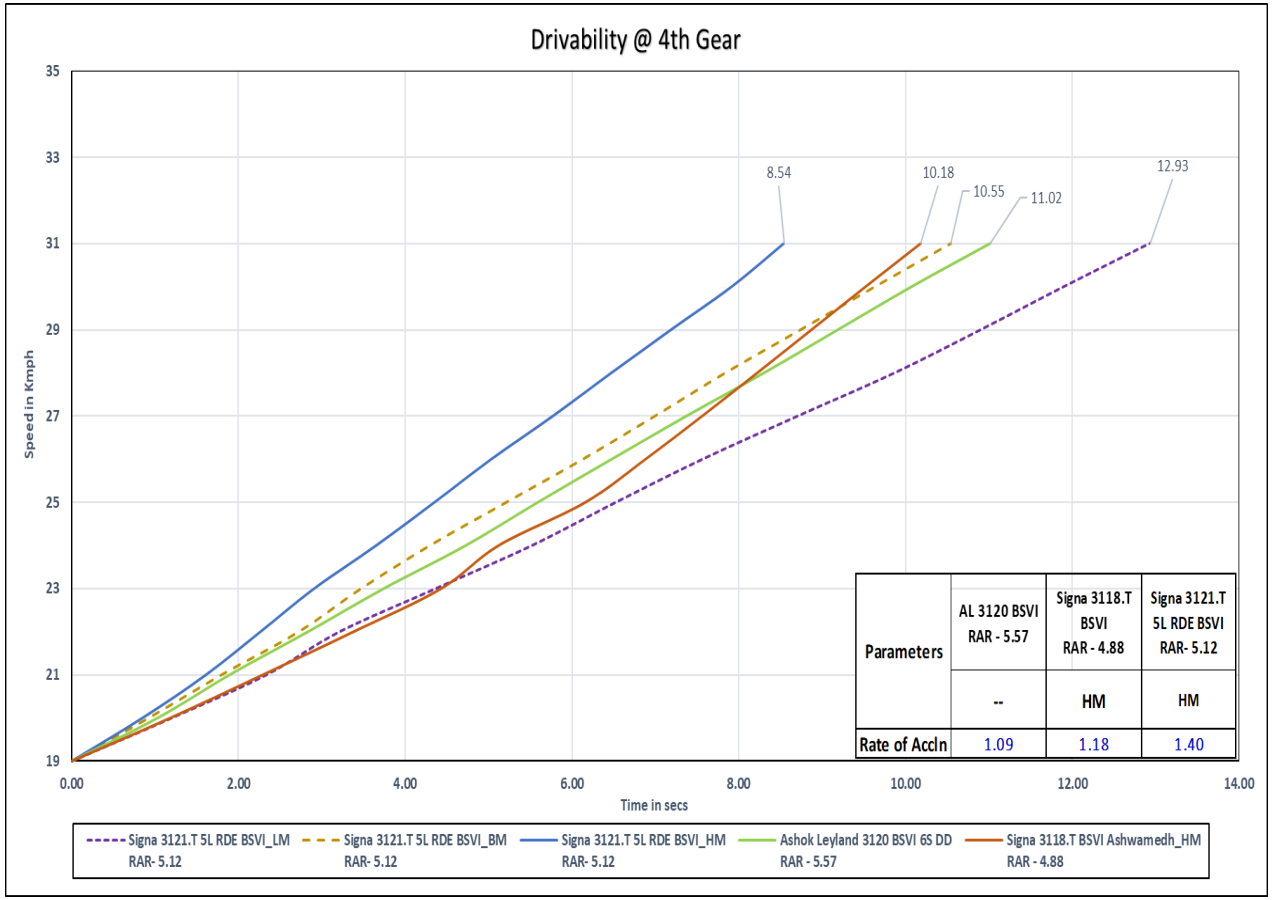


Parameters	AL 3120 BSVI RAR - 5.57	Signa 3118.T BSVI RAR - 4.88	Signa 3121.T 5L RDE BSVI RAR- 5.12
	--	HM	HM
Rate of Accln	1.56	2.08	2.86

- In 1st & 2nd gear, WOT Drivability of Signa 3121.T 5L Cx BSVI-II vehicle is better as compared with AL 3120 & Signa 3118.T Ash. BSVI-I .

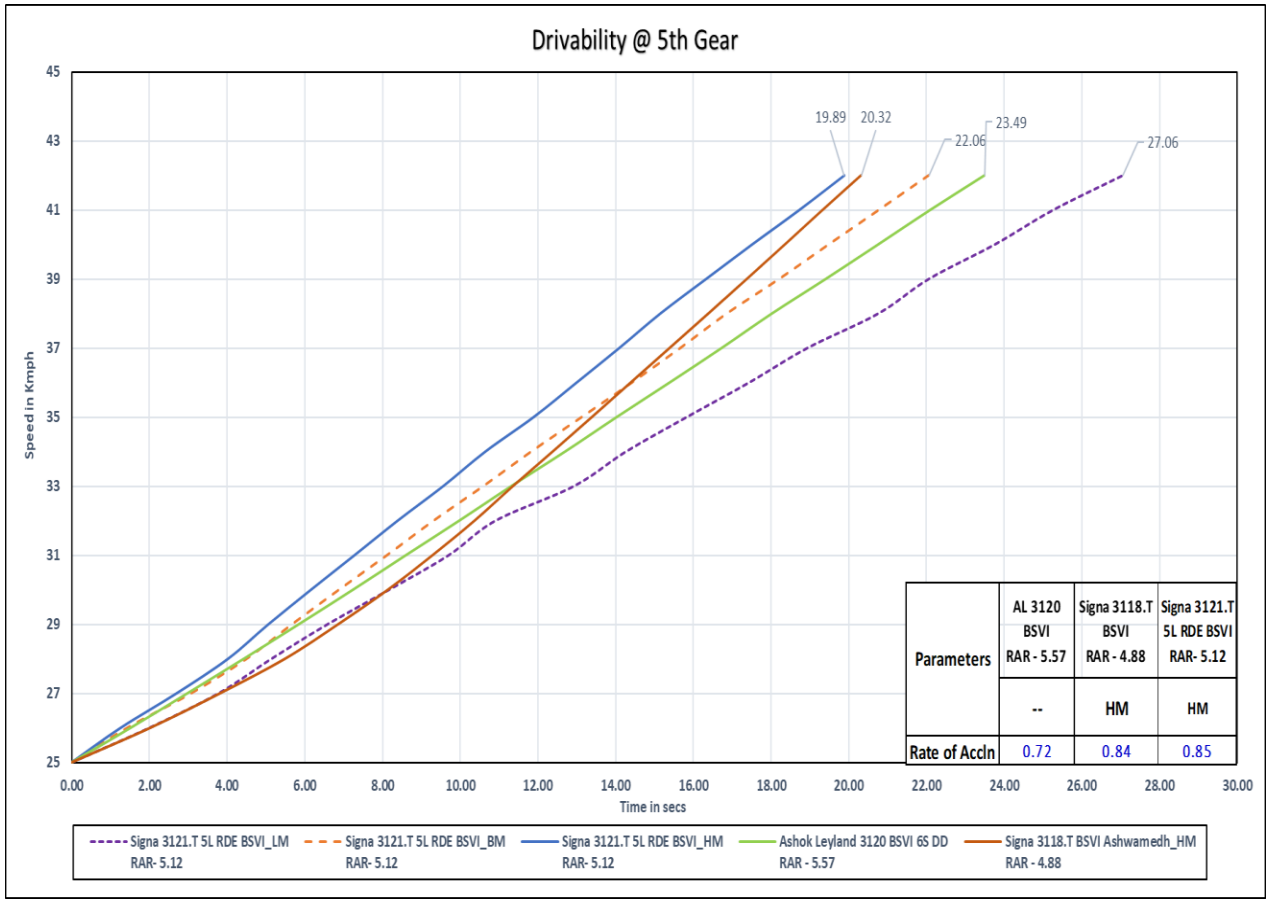
WOT Drivability @ 3rd & 4th Gear

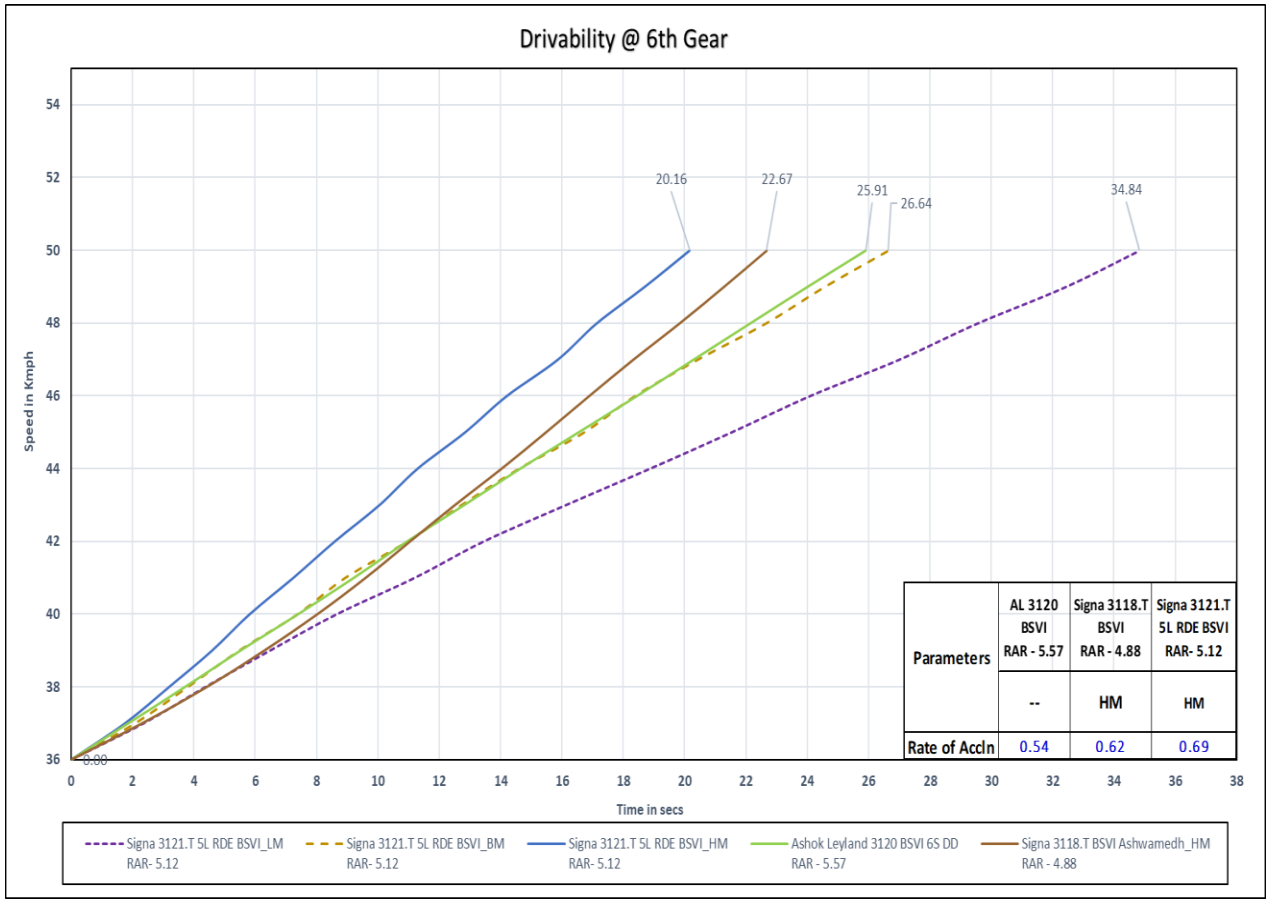




➢ In 3rd & 4th gear, WOT Drivability of Signa 3121.T 5L Cx BSVI-II vehicle is better as compared with AL 3120 BSVI & Signa 3118.T Ash. BSVI-I

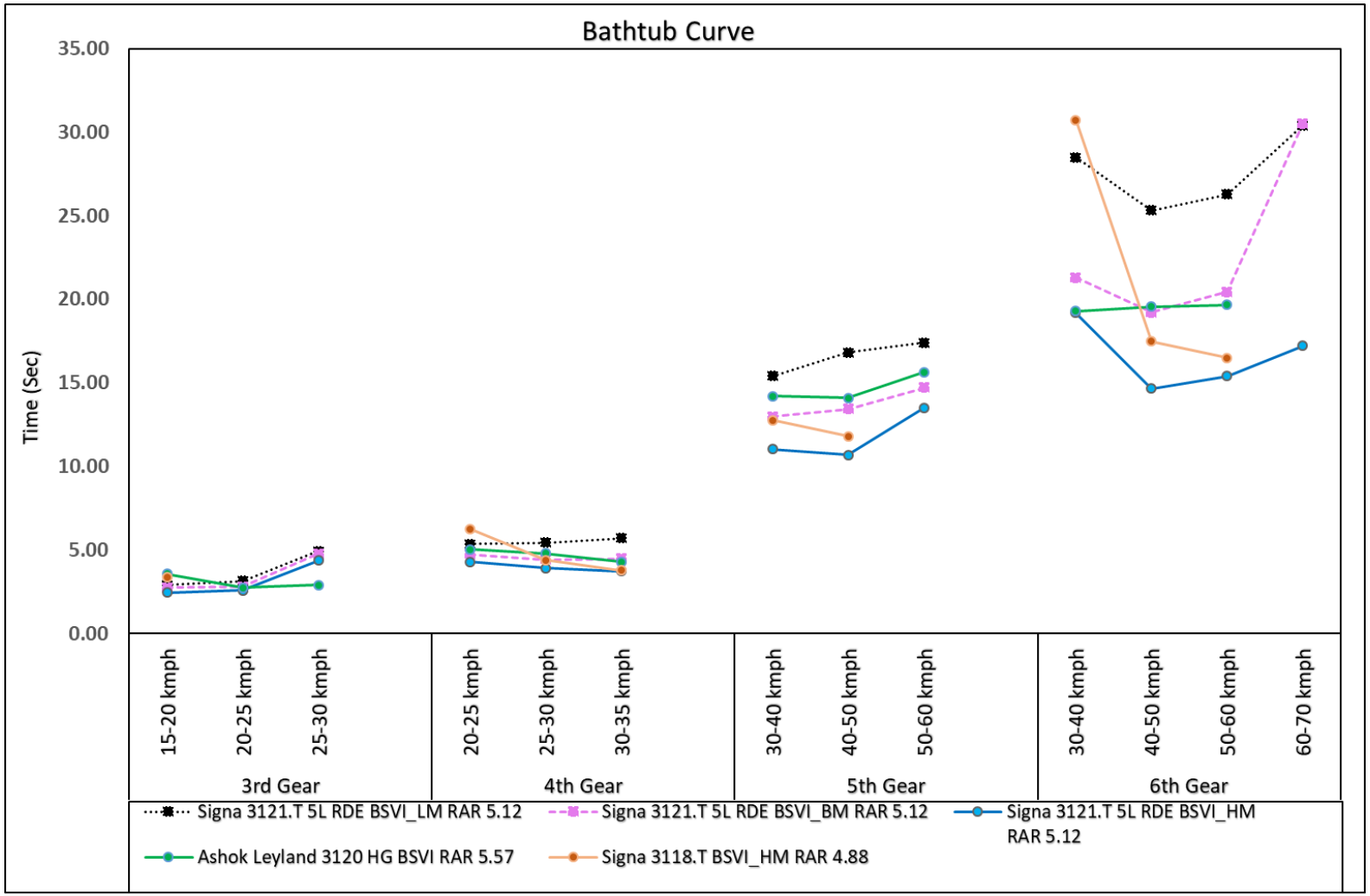
WOT Drivability @ 5th & 6th Gear





➢ In 5th & 6th gear, WOT Drivability of Signa 3121.T 5L Cx BSVI-II vehicle is better as compared with AL 3120 BSVI & Signa 3118.T Ash. BSVI-I

Bathtub Drivability Comparison with Benchmark & Base Vehicles



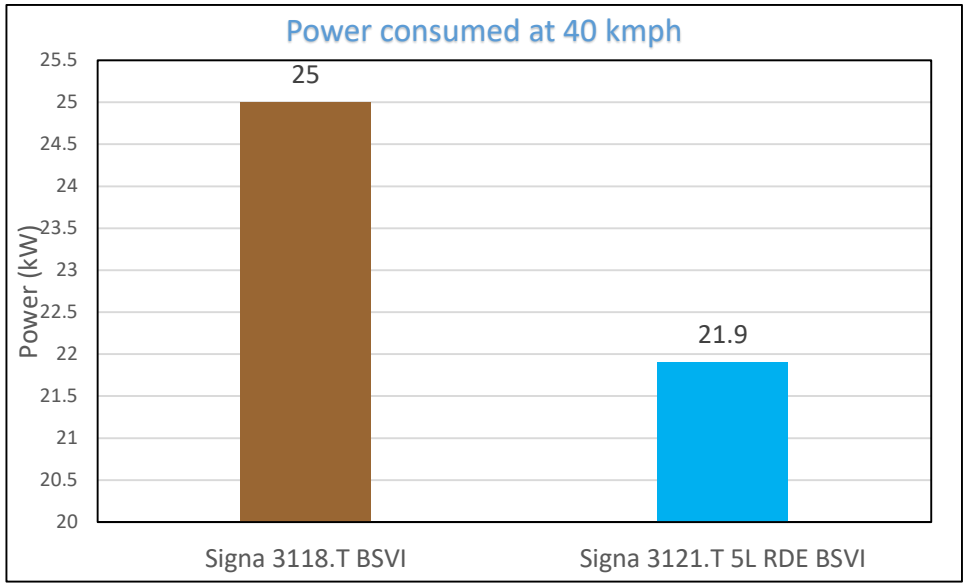
Gear	Speed band	Ashok Leyland 3120 HG BSVI RAR 5.57	Signa 3118.T BSVI_HM RAR 4.88	Signa 3121.T 5L RDE BSVI_LM RAR 5.12	Signa 3121.T 5L RDE BSVI_BM RAR 5.12	Signa 3121.T 5L RDE BSVI_HM RAR 5.12
		Time in Sec	Time in Sec	Time in Sec	Time in Sec	Time in Sec
3rd Gear	15-20 kmph	3.59	3.38	2.91	2.76	2.45
	20-25 kmph	2.75		3.15	2.80	2.58
	25-30 kmph	2.90		4.93	4.77	4.37
4th Gear	20-25 kmph	5.04	6.27	5.37	4.74	4.29
	25-30 kmph	4.79	4.40	5.44	4.44	3.91
	30-35 kmph	4.31	3.79	5.71	4.47	3.73
5th Gear	30-40 kmph	14.20	12.77	15.41	12.99	11.03
	40-50 kmph	14.11	11.81	16.83	13.45	10.70
	50-60 kmph	15.66		17.41	14.72	13.50
6th Gear	30-40 kmph	19.31	30.73	28.48	21.31	19.19
	40-50 kmph	19.57	17.48	25.32	19.21	14.65
	50-60 kmph	19.67	16.49	26.28	20.44	15.42
	60-70 kmph	22.28		30.41	30.50	17.23

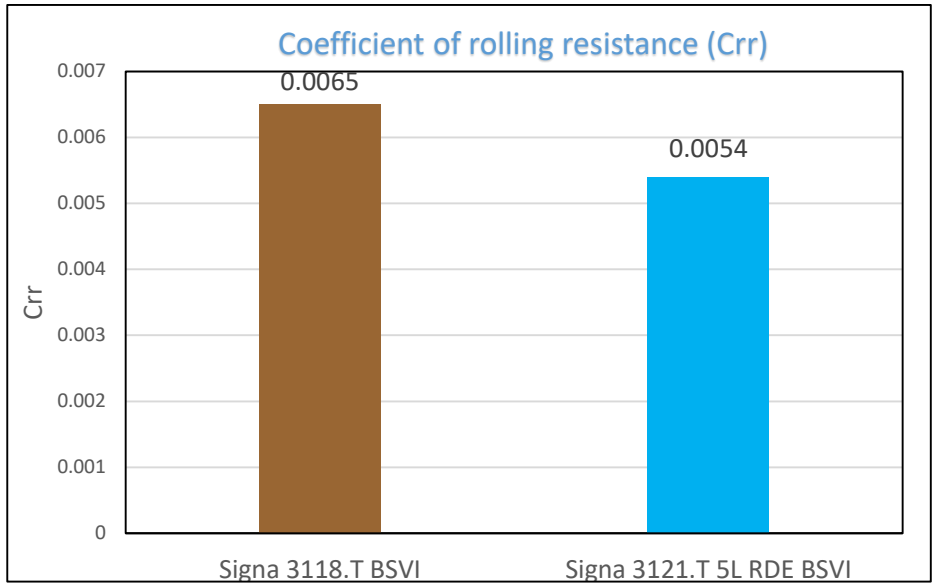
➢ Bathtub Drivability of Signa 3121.T 5L Cx BSVI-II is better as compared with AL 3120 BSVI & Signa 3118.T Ash. BSVI-I vehicle & meetings PAL target of "C", except in 3rd gear (25-30kmph) speed.

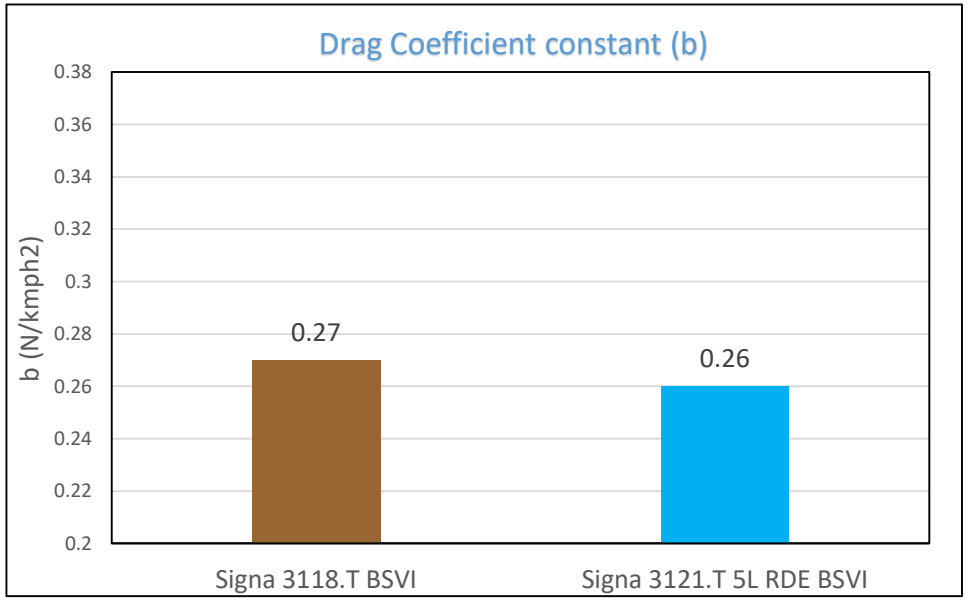
Coast Down Comparison with Benchmark & Base Vehicles

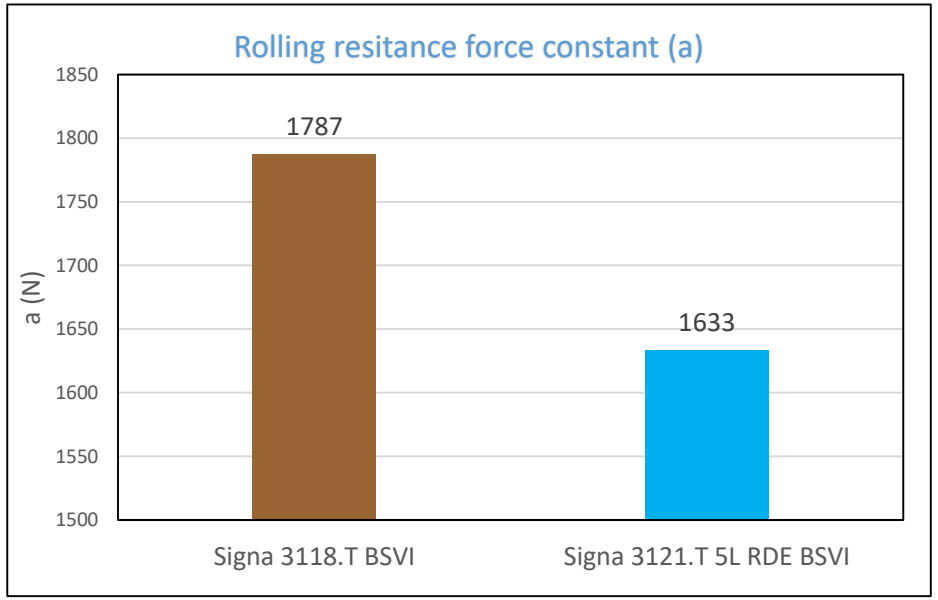
TATA MOTORS
Connecting Aspirations

Vehicle	Power consumed at 40 kmph	a	b	Coeff of rolling resistance
	kW	N	$N/kmph^2$	--
Signa 3118.T BSVI	25	1787	0.27	0.0065
Signa 3121.T 5L RDE BSVI	21.9	1633	0.26	0.0054

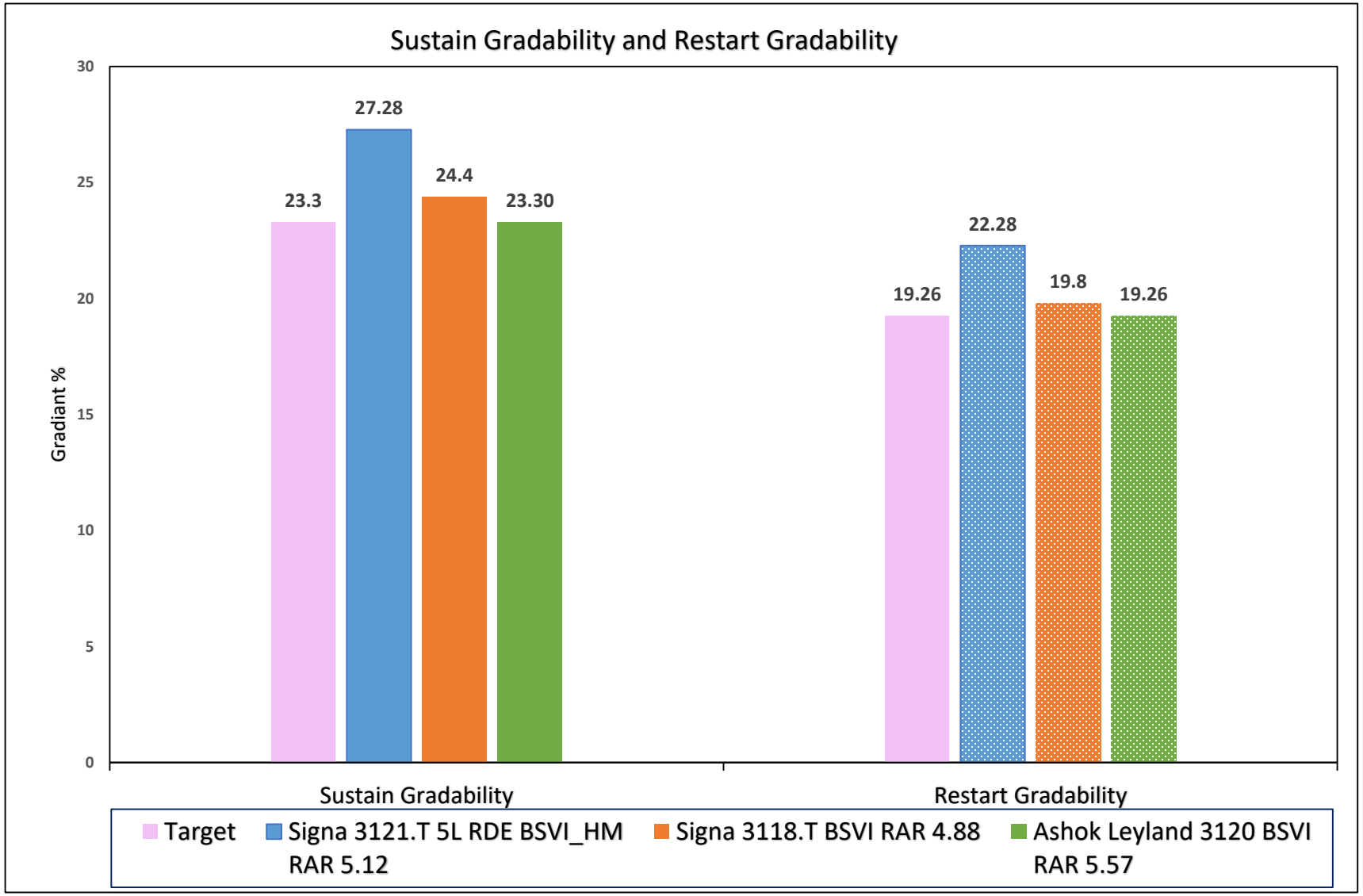








Sustained and Restart Gradeability



Test	Target	Ashok Leyland 3120 BSVI RAR 5.57	Signa 3118.T BSVI RAR 4.88	Signa 3121.T 5L RDE BSVI_HM RAR 5.12
Sustain Gradability	23.3	23.30	24.4	27.28
Restart Gradability	19.26	19.26	19.8	22.28

Thank You!

TEST REQUISITION

Ref.		Date:	02/04/2023

To,
ERC Testing
Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3121.T BSVI				
2	Part Number	51800151R,				
3	Drawing number	518000000005,	Mod status	E,2,	Mod Date	20/04/2023
4	Vehicle Model / Project	TATA LPT 3121 5L RDe BSVI				
5	WBS Number	P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T				
6	Supplier	NA				
7	Number of samples					
8	Inspection report	Enclosed	☑ Yes	☐ No	If No commitment date:	
9	Material report	Enclosed	☑ Yes	☐ No	If No commitment date:	
10	Suppliers test report	Enclosed	☑ Yes	☐ No	If No commitment date:	
11	Reason for testing	First Source Development		☐	Weight Reduction	☐
		Alternate source development		☐	Cost Reduction	☐
		Data generation Test		☐	Regulatory test	☐
		Any other (pl. specify)		☐	New Development	☐
12	What testing required?	DYNAMIC				
13	Remarks	TATA SIGNA 3121.T BSVI				
14	Peripheral Parts Availability	YES				

Authorized Signature:	
Name:	
Dept:	

Contact Person:	
Phone Number	

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

TATA MOTORS LIMITED ERC PUNE	TEST PLAN VEHICLE TESTING DEPARTMENT	DATE: 24/07/2023			
Description: Performance evaluation of Signa 3121.T Cx RDE in comparison with Ashok Leyland 2820 BSVI.		**Test Item No:** 42587			
		New	✔	**Moderate**	✘
		Major	✘	**Minor**	✘

Part No: 51800151R, **Drawing No:** 518000000005,	**Modification No:** E,2,	**Enclosures:** Annexure 1 - 6
Make: NA	**WBS Number:** P.22.EA.501 -RDE Truck 5.6 Ltr 3123.T / 3523.T	**Copies to** :
Vehicle Type: TATA LPT 3121 5L RDe BSVI	**No. of Samples:** 1	

Test Requested by:	VEHICLE-I	**Equipments:** Signa 3121.T Cx RDE
Test Requisition Reference:	Testing-Outdoor/15087	
Reason for Test/s:	Component & System Level Performance Tests	
Regulatory Requirement:	a) Component: As per PED standards	
	b) Vehicle: As per PED standards	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Performance & Driveability tests	As per PED standards	As per PALS	1	Palash Das	Beta	done

REMARKS: Test plan for test execution	**INTERNAL REF:**	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	**CHECKED BY**	**APPROVED BY**

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